lit-tag-viewer User Guide



August 28, 2026

1 Overview

The `lit-tag-viewer` module of the `lit-tag` app allows users to search, visualize, and summarize a tagged literature database created using the lit-tag-builder module.

The **Search database** feature enables filtering based on paper criteria (*author, publication year, title*), notes criteria, and tag criteria (*user-defined categories*). Papers that meet the selected criteria are displayed in a table and can be exported to `.csv`, `.bib`, or `.ris` files.

The **Summary plot** feature supports visualization of the full or filtered database. The app generates a bar graph displaying the number of papers (y-axis) for any paper or tag criteria (x-axis). Users also have the option to create a stacked bar graph with a color key to visualize two categorical variables.

The **Summary table** feature supports the creation and export of tables (`.csv`) with user selected fields from the lit-tag database.

The **Reports** feature generates a downloadable file (`.html`, `.pdf`, `.docx`) that summarizes the full or filtered database. In addition to basic bibliographic information, reports can include the paper abstract, tags, and notes.

The `lit-tag-builder` is a separate lit-tag module for creating a lit-tag database that can be viewed with the lit-tag-viewer.

Recommended citation for lit-tag: McElhany, P., Grabb, K. C., Wood, M. M. (2025). "lit-tag: An app for adding custom tags and notes to a citation database".

2 Using lit-tag-viewer

2.1 Load database

Before using `lit-tag-viewer`, two files must be loaded: 1) a **lit-tag database file** (`.csv`)made using the lit-tag-input app and 2) a **lit-tag tag categories file** (`.xlsx`) that contains tag categories, tag names and tag value options.

2.1.0.1 Embedd database

Users may interact with versions of `lit-tag-viewer` with and embedded database `.csv` file and categories `.xlsx` files that were created for a specific project. To load and embedded database, simply click **Load Database** button (Figure 1). A table with the authors, year and title of all the references in the database will appear on the right side of the window. You can download the embedded database `.csv` and categories `.xlsx` file by clicking **Download database** and **Download tag categories**, respectively.



Figure 1: Load Database tab for an embedded database.

2.1.0.2 User-provided files

More commonly, users may utilize the `lit-tag-viewer` module to view database and category files created by by themself or someone else with `lit-tag-builder` and saved locally on their computer. To load the database, click browse and select the appropriate `.csv` file on your local hard drive (Figure 2). To load the categories, click browse and select the appropriate `.xlsx` file on your local hard drive. The creation and contents of these files are described in the *lit-tag-builder User Guide* accessible in the `lit-tag-builder` module. After the database and categories files have been selected, click the **Load database** button. A table with the authors, year and title of all the references in the database will appear on the right side of the window.

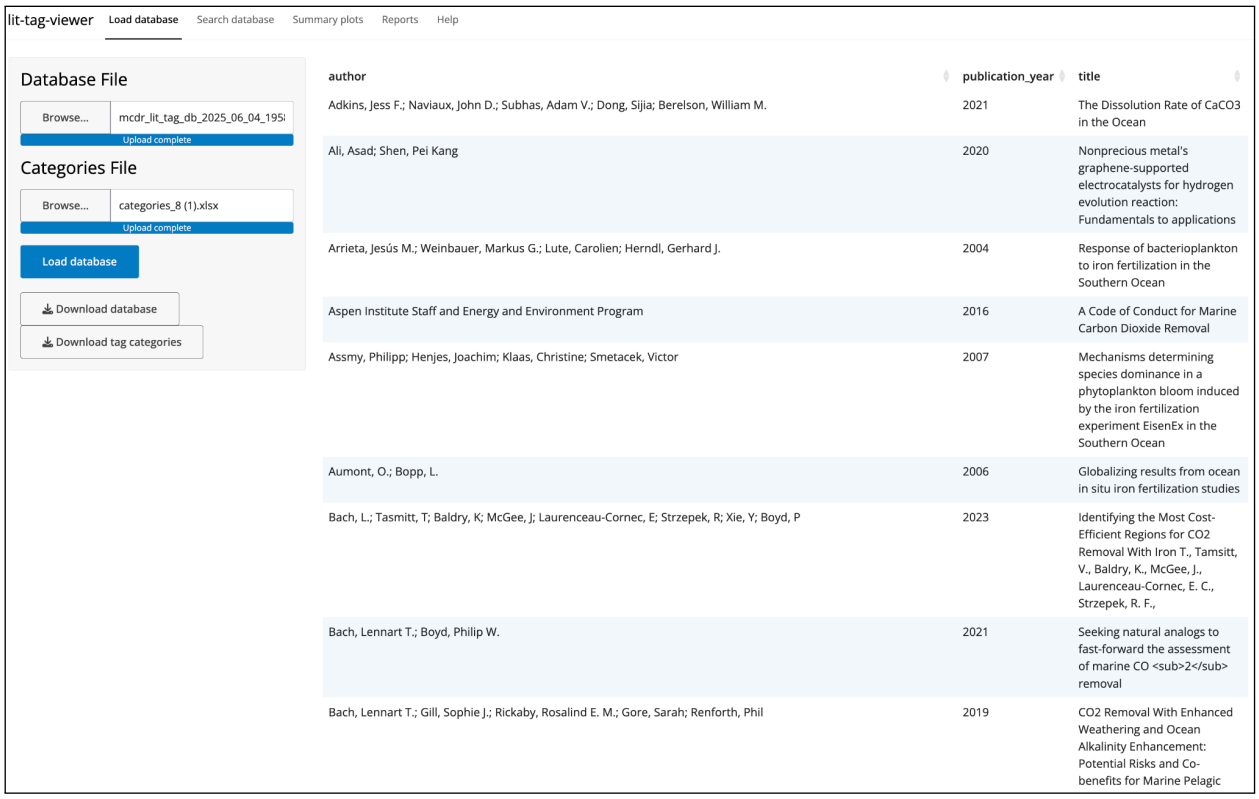


Figure 2: Load database tab with an example database loaded.

2.2 Search database

The **Search database** tab can be used to filter the full database to identify a subset of references that meet user-selected criteria (Figure 3).

The left side of the window displays the **Paper table (filtered)**. Before filtering criteria

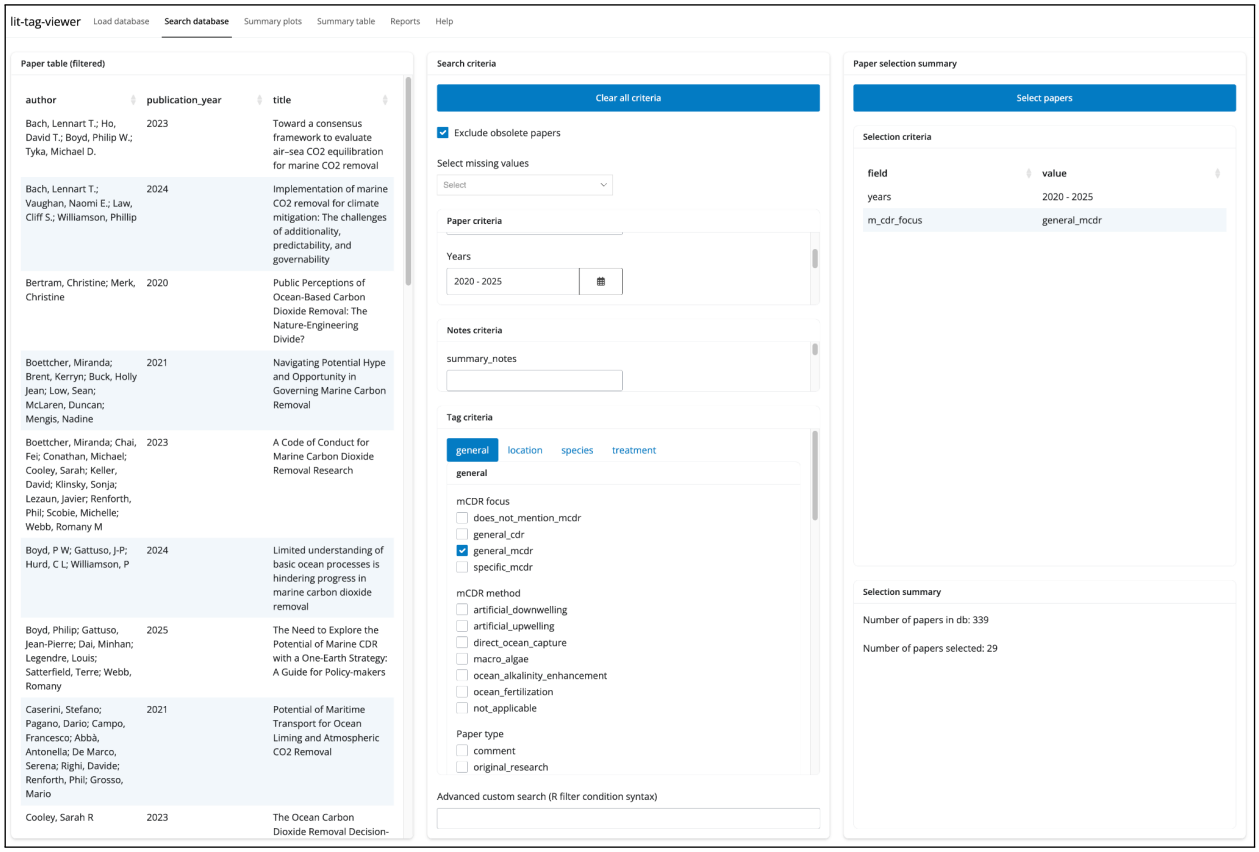


Figure 3: Example screen shot of database search.

are supplied, the full database will be displayed. After filtering, the table will be updated to display only those papers that meet the filtering criteria. The center of the window displays the search criteria that may be applied by the user. The right of the window displays the selection criteria that have been applied, the number of papers meeting those criteria, and the export functionality for saving filtered databases.

To filter papers, enter search terms and/or select criteria check boxes, then click the **Select papers** button. Clicking **Clear criteria** will reset search criteria.

Filtered databases may be downloaded as `.csv`, `.bib`, or `.ris` files by clicking the appropriate **Export selection** button after entering a field name in the **Export filename** text field.

2.2.0.1 Filter by Paper criteria

Users may search by author name or keywords in the title or abstract. Strings of keywords are allowed, and multiple search terms should be separated by a semi-colon (`;`). Search strings are case sensitive.

2.2.0.2 Filter by Notes criteria

Users may search by keywords in notes provided by the database reviewer. Strings of keywords are allowed, and multiple search terms should be separated by a semi-colon (`;`). Search strings are case sensitive.

2.2.0.3 Filter by Tag criteria

Users may filter by one or more tag categories by selecting checkboxes associated with tag values. The tag categories and values are those supplied in the **lit-tag tag categories file** (`.xlsx`).

2.2.0.4 Advanced custom search

More complicated searches of the database can be done by entering an advanced custom search. The syntax for the search is that used in the R tidyverse "filter" function. All the base R and tidyverse functions are available for search. Searches may often want to use the "str_detect" function.

2.2.0.5 View tags on individual papers

In the search tab, selecting a paper from the table of filtered papers in the left panel will bring up a window with the tags, notes and abstract specific to that paper (Figure 3). To close the tab, click **Dismiss** at the bottom.

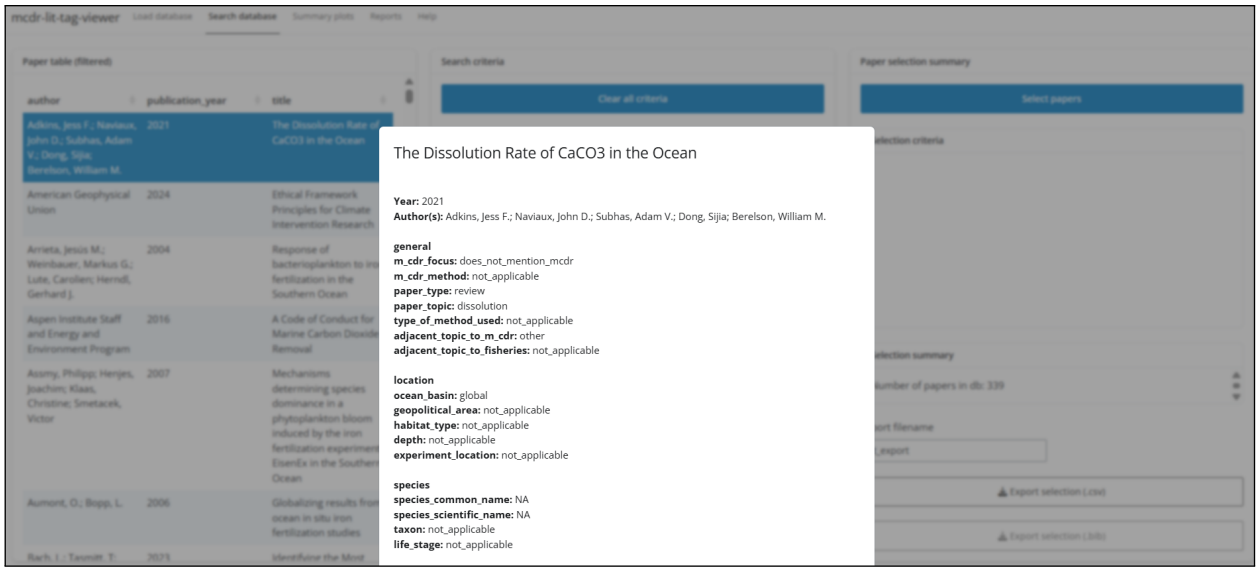


Figure 4: Pop up summary of tags on selected paper.

2.3 Summary plots

The **Summary plots** tab is used to generate paper frequency bar graphs, stacked frequency bar graphs and histograms for bibliographic information and tag categories. The user may select **Full dataset** to plot data for the entire database, or may select **Filtered database** to plot only the filtered database currently selected in the "Search database" tab. To change the filtered set, use the **Database search** tab.

2.3.1 Frequency bar plots

Plots are generated by selecting an x-axis variable in the **Plot x-axis variable** drop down menu (Figure 5). A stack variable may also be selected under **Plot stack variable** to visualize two categorical variables together (Figure 6). Percentage or number of papers represented by each bar may be added by making the appropriate selection under **Show bar data value**. Users may optionally exclude **not applicable** or **missing** values from the plot.

When plotting stacked variables, users may also opt to **Combine multi-select** for the x-axis and/or stack variable. This option can be applied when, for example, the stacked variable has multiple selections for each x-axis variable. The default plot will show the proportion of each stack value, not the absolute count. Applying **Combine stacked** will instead show the absolute count and separate stack variables for each combination of multiple selections.

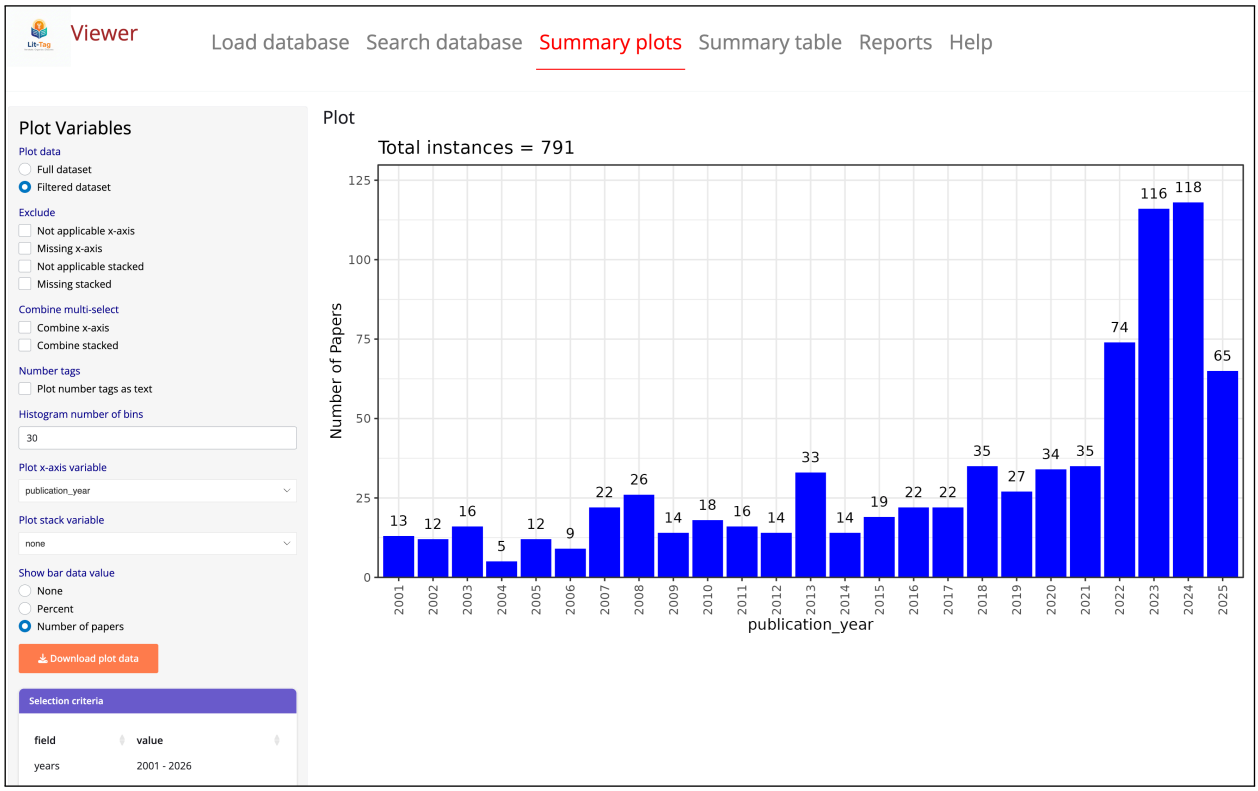


Figure 5: Example frequency bar plot for filtered data.

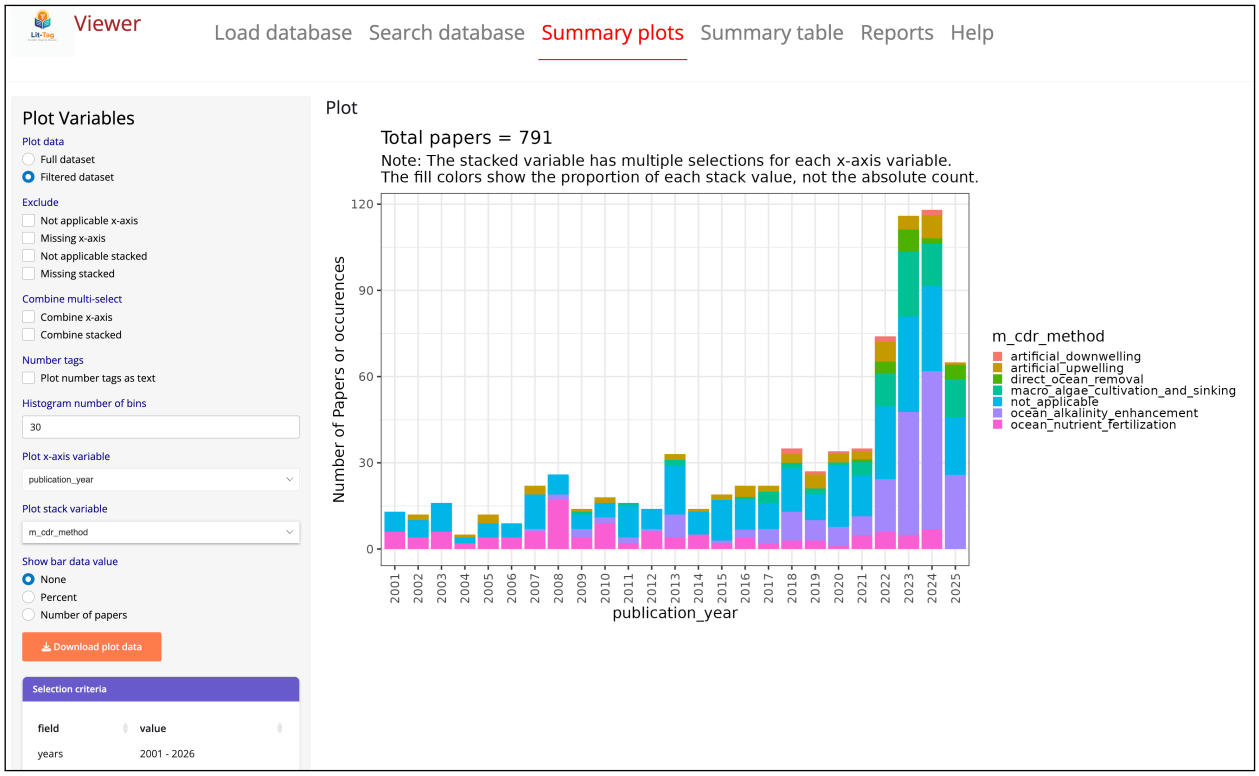


Figure 6: Example of stacked bar plot.

2.3.2 Number tag plots

By default, number tags are displayed as a histogram (Figure 7) with the **Histogram number of bins** input controlling how many bins the tag number range is divided into for plotting. Optionally, selecting **Plot number tags as text** will display number tags as frequency bar or stacked bar plots.

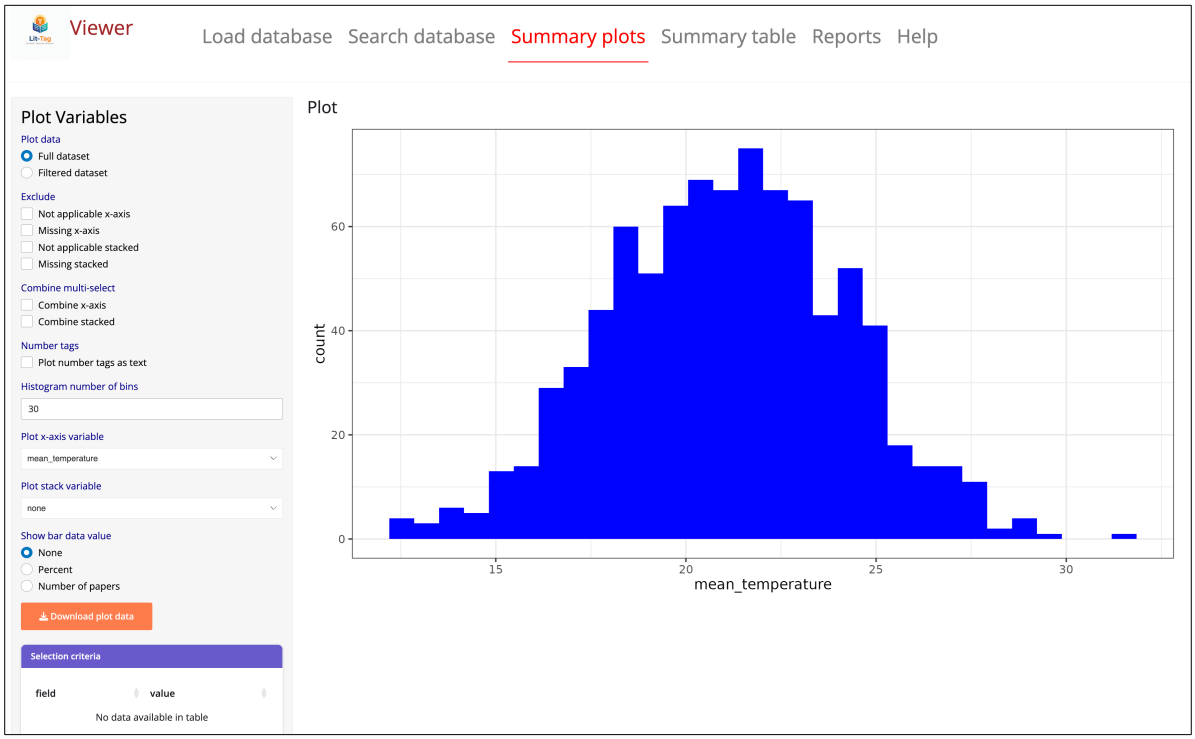


Figure 7: Example histogram for number tag.

2.4 Summary table

A summary table can be created on the **Summary table** tab with user selected fields showing either the complete data set or the filtered data set (Figure 8). The column order can be changed by clicking and dragging the column header. The table can be sorted by clicking on the header of the column to be used for sorting. The summary table can be downloaded as a `.csv` file.

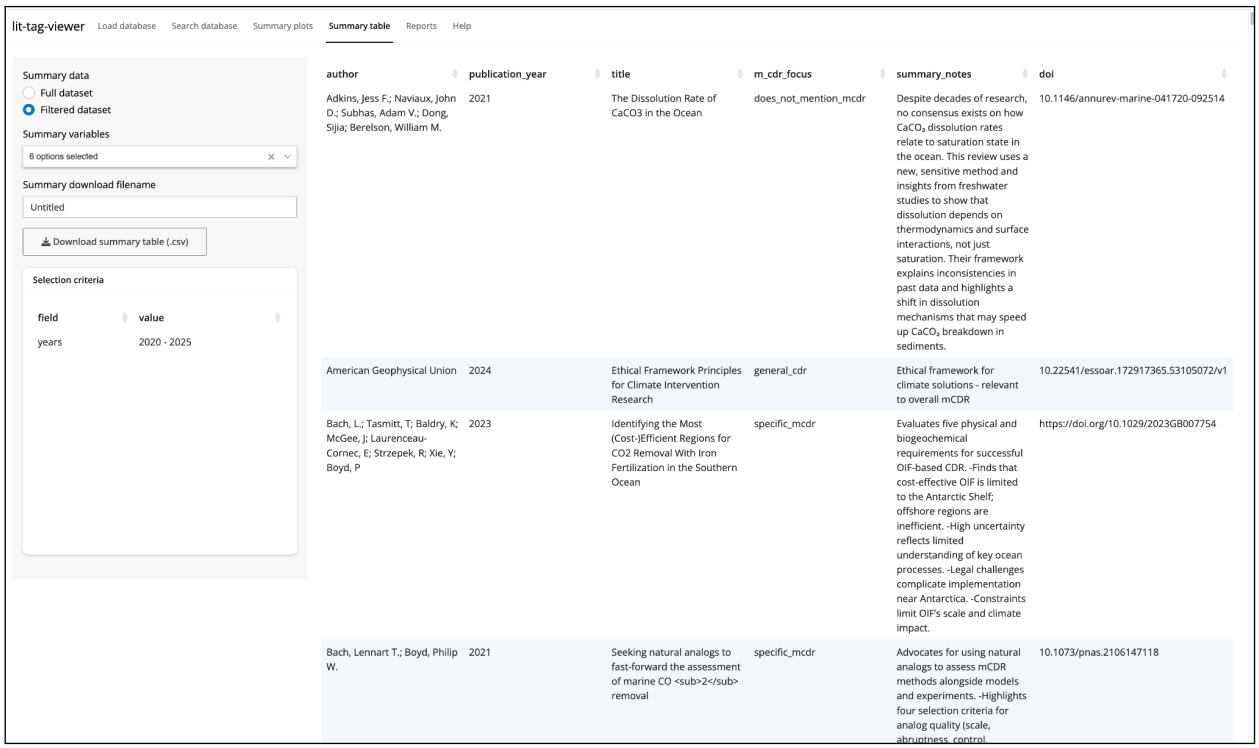


Figure 8: Example summary table.

2.5 Reports

The **Reports** tab allows users to generate and output summary reports for the full database or the filtered database currently selected in the **Search database** tab (Figure 9).

Users may enter a **Report title** and **Report author**, and opt to sort the report by author, publication year, or title. The default settings produce a report that includes basic bibliographic information, paper_url, abstract, tags, missing_tags, not_applicable_tags, notes, and pagebreaks. Users may customize the report contents by selecting only desired elements.

Click the **Show report** button to view the report on the right side of the window.

To save the report to your computer as an `.html`, `.pdf`, or `.docx` file, edit **Report filename** and select your preferred file format. Then click the **Download report** button.



Figure 9: Example report